

Jason Xie

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EXPERIENCE

Software Developer – Internship

May 2026 – Aug 2026

Vertex Inc.

Greater Philadelphia

- Benchmarked **Jedis** vs. **Lettuce** Java Redis clients to guide production client selection for a distributed caching layer on **AWS ElastiCache**, building **JMH** microbenchmarks against **Dockerized** Redis cluster topologies and **Terraform/EC2** test environments with scoped **IAM** roles for in-VPC access.
- Measured latency, throughput, GC pressure, connection pooling, and multi-AZ replica routing. Found Lettuce shared **1.9×** faster under 200-thread concurrency with **7.6×** lower write allocation and **~15%** faster writes.
- Modeled throughput scalability from **k6** load tests by fitting **Universal Scalability Law** curves and validating concurrency with **Little's Law**, isolating bottlenecks across **Datadog APM**, **CloudWatch**, and **Aurora** metrics.
- Refreshed cloud performance baselines across **12** hosting levels spanning **L1**, **L2**, and **L3** tiers, benchmarking both **REST** and **SOAP** APIs to quantify up to **~75%**, **~74%**, and **~50%** uplift over prior targets and translating results into customer-facing capacity guidance for a tax platform used by **>60% of the Fortune 500**.

Co-Founder & CTO

2025 – Present

Scrap-It

Hamilton, ON

- Architected and shipped a published cross-platform photo-sharing app in **React Native**, **Expo**, and **TypeScript**, backed by **Supabase/PostgreSQL**, **Clerk** authentication, and **Cloudflare R2** object storage, with the marketing site deployed on **Cloudflare Pages**.
- Engineered secure backend infrastructure using PostgreSQL **Row-Level Security**, storage access policies, album-level permissions, and moderation, reporting, and user-blocking flows.
- Published to the **Apple App Store** and **Google Play**, managing production QA, store review and privacy compliance for a user-generated-content app, and a launch campaign generating **10,000+** social media views and **2,000+** site visits.

AI Backend/Infrastructure Developer – Internship

Mar 2025 – May 2025

Atlantis AI

Remote

- Shipped a Firefox extension integrating a lightweight **LLM** into the browsing workflow.
- Evaluated and tuned models and prompts to improve output quality while lowering token costs.
- Built UI with **React (TypeScript)** + **Tailwind**; integrated LLM via Hugging Face **Inference API**.

Front-End Engineering – Internship

Feb 2025 – Apr 2025

Senticik (AI Consulting Firm)

Vancouver, BC

- Built a responsive site with client-side validation using **HTML**, **CSS**, and **JavaScript**.
- Set up **Git** workflow and cloud deployment with handoff documentation for client maintenance.

PROJECTS

Remind-Bot (Chrome Extension) | *Chrome MV3, JavaScript, HTML/CSS, Chrome Alarms/Notifications APIs*

- Implemented per-reminder alarms, actionable notifications, and automatic rescheduling on browser storage changes.
- Found users through social-first **product messaging** and short-form video content: **~3k views** → **~11 installs**.

Eat-Safe Toronto | *Python, pandas, SQLite, JavaScript, HTML/CSS, Papa Parse, GitHub Pages*

- Public site to look up Toronto restaurant health inspections by name/address; surfaces latest status and full history to inform dining choices.
- Normalized **132k** rows to **SQLite** and published a browser-parsable CSV; indexed lookups return in under **1s**.

EDUCATION

Bachelor of Applied Science in Computer Science

Hamilton, ON

McMaster University

Sept. 2024 – May 2028

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C, C++, C#, SQL, HTML/CSS, HCL (Terraform)

Frameworks & APIs: React, Next.js, React Native, Expo, FastAPI, Tailwind CSS, REST APIs, SOAP APIs

Cloud & Infrastructure: AWS (EC2, ElastiCache, Aurora, CloudWatch, IAM, VPC/Security Groups) via both AWS Console and **Terraform** IaC, Cloudflare (R2, Pages, Workers), Docker, Redis, Supabase, Render

Data & Tooling: PostgreSQL, SQLite, Prisma, k6, JMH, Maven, Datadog APM, Git, GitHub Actions